

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network
Security **COURSE CODE:** CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Kanmani L(192511414) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A small, square, grayscale image showing a handwritten signature in blue ink on a light-colored background. The signature appears to be 'Kanmani L.'.

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.

- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Scenario 1: Event Communication Security

Title: Event Communication Security Using Caesar Cipher (Shift = 4)

Introduction

In any organization, confidential information should only be accessible to authorized members. One of the simplest classical encryption techniques used for secure communication is the Caesar Cipher. It is a substitution cipher where each letter in the plaintext is shifted by a fixed number of positions in the alphabet. In this scenario, the message "MEET AT NOON" is encrypted using a shift value of 4 before being shared with the student organization's core members.

Given Data

Particular	Value
Plaintext	MEET AT NOON
Encryption Technique	Caesar Cipher
Shift Value	+4
Decryption Shift	-4

Theory

The **Caesar Cipher** is a classical substitution cipher in which every letter of the original message is replaced by another letter located a fixed number of positions away in the alphabet.

Encryption Process

Plaintext → Shift each letter by +4 → Ciphertext

Decryption Process

Ciphertext → Shift each letter by -4 → Original Plaintext

(a) Encryption of the Message

Step 1: Original Message

MEET AT NOON

Step 2: Apply a Shift of +4

Each alphabet is shifted **4 positions forward**.

Example:

M → Q

E → I

T → X

A → E

N → R

O → S

Step 3: Encryption Table

Plain Letter Shift (+4) Cipher Letter

M → Q

E → I

E → I

T → X

Space → Space

A → E

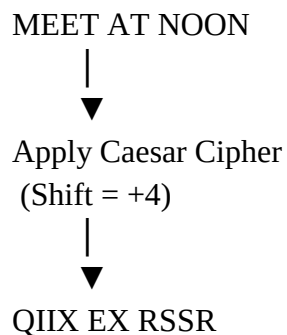
Plain Letter Shift (+4) Cipher Letter

T	→	X
Space	→	Space
N	→	R
O	→	S
O	→	S
N	→	R

Encrypted Message

Plaintext	Ciphertext
MEET AT NOON	QIIX EX RSSR

Encryption Flow



(b) Decryption Process

The authorized member knows that the message has been encrypted using a **shift value of 4**. To recover the original message, each encrypted letter is shifted **4 positions backward**.

Decryption Table

Cipher Letter Shift (−4) Plain Letter

Q	→	M
---	---	---

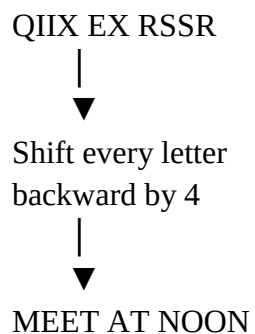
Cipher Letter Shift (-4) Plain Letter

I	→	E
I	→	E
X	→	T
Space	→	Space
E	→	A
X	→	T
Space	→	Space
R	→	N
S	→	O
S	→	O
R	→	N

Recovered Message

Ciphertext	Recovered Plaintext
QIIX EX RSSR	MEET AT NOON

Decryption Flow



Explanation

The sender encrypts the confidential message by shifting every alphabet **4 positions forward**, producing the ciphertext "**QIIX EX RSSR**". Since the authorized receiver knows the same shift value, they simply shift each encrypted letter **4 positions backward** to retrieve the original message.

This method ensures that only users who know the encoding rule can understand the message. Although the Caesar Cipher provides only basic security and can be easily broken using brute-force attacks, it effectively demonstrates the fundamental concepts of encryption and decryption.

Conclusion

The message "**MEET AT NOON**" is successfully encrypted as "**QIIX EX RSSR**" using the **Caesar Cipher** with a **shift value of 4**. An authorized member can retrieve the original message by applying a **4-position backward shift**. This example illustrates how classical encryption techniques can be used to protect confidential information during communication.

Result

Plaintext → **MEET AT NOON**

↓

Caesar Cipher (Shift +4)

↓

Ciphertext → **QIIX EX RSSR**

↓

Decryption (Shift -4)

↓

Recovered Message → **MEET AT NOON**

Scenario 2: Intercepted Suspicious Message

Title: Analysis of an Intercepted Message Using the Atbash Cipher

Introduction

During network monitoring, a cybersecurity intern intercepts the coded message "**GSRH RH Z HVXIVG**". Since the message appears to use a simple substitution pattern rather than a complex encryption algorithm, it is likely encrypted using a classical cipher. One of the most common substitution ciphers with this pattern is the **Atbash Cipher**, where each letter is replaced by its opposite letter in the alphabet ($A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$, etc.).

Given Data

Particular	Value
Ciphertext	GSRH RH Z HVXIVG
Suspected Cipher	Atbash Cipher
Cipher Type	Monoalphabetic Substitution Cipher

Theory

The **Atbash Cipher** is a classical substitution cipher that uses a **reversed alphabet** for encryption and decryption.

Alphabet Mapping

Normal Alphabet	A B C D E F G H I J K L M
Atbash Alphabet	Z Y X W V U T S R Q P O N
Normal Alphabet	N O P Q R S T U V W X Y Z
Atbash Alphabet	M L K J I H G F E D C B A

Decryption Process

Ciphertext → Reverse Alphabet Mapping → Original Message

(a) Deduce the Original Message

Step 1: Intercepted Ciphertext

GSRH RH Z HVXIVG

Step 2: Apply Atbash Decryption

Each letter is replaced by its opposite letter in the alphabet.

Example:

G → T

S → H

R → I

H → S

Z → A

Step 3: Decryption Table

Cipher Letter Opposite Letter Plain Letter

G	→	T
S	→	H
R	→	I
H	→	S
Space	→	Space
R	→	I
H	→	S

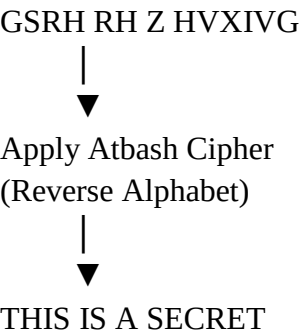
Cipher Letter Opposite Letter Plain Letter

Space	→	Space
Z	→	A
Space	→	Space
H	→	S
V	→	E
X	→	C
I	→	R
V	→	E
G	→	T

Original Message

Ciphertext	Recovered Plaintext
GSRH RH Z HVXIVG	THIS IS A SECRET

Decryption Flow



(b) Justification of the Answer

The intercepted message is most likely encrypted using the **Atbash Cipher** because of its distinct substitution pattern.

Pattern Analysis

Observation	Analysis
The word RH appears twice	It decrypts to IS , a common English word.
The single-letter word Z	Decrypts to A , which is a valid one-letter English word.
The word HVXIVG	Decrypts to SECRET , forming a meaningful sentence.
The complete sentence	Produces the grammatically correct phrase " THIS IS A SECRET. "

Characteristics of the Atbash Cipher

Feature	Description
Cipher Type	Monoalphabetic Substitution Cipher
Key Required	No
Encryption Rule	Reverse the alphabet ($A \leftrightarrow Z$, $B \leftrightarrow Y$, etc.)
Decryption Method	Apply the same reverse mapping
Spaces	Remain unchanged

=

Why This Cipher is the Most Likely Choice

- The repeated words follow meaningful English language patterns.
- The single-letter word "**Z**" correctly decrypts to "**A**".
- Every encrypted letter follows the fixed reverse alphabet mapping.
- The decrypted sentence is grammatically correct and meaningful.

These characteristics strongly indicate that the message was encrypted using the **Atbash Cipher**.

Explanation

The Atbash Cipher uses a **fixed reverse alphabet substitution**, making encryption and decryption identical. When the intercepted message "**GSRH RH Z HVXIVG**" is decoded using this method, it produces "**THIS IS A SECRET.**" By analyzing repeated words, letter patterns, and the reverse alphabet mapping, investigators can confidently determine both the original message and the encryption technique used.

Conclusion

The intercepted ciphertext "**GSRH RH Z HVXIVG**" is successfully decrypted as "**THIS IS A SECRET**" using the **Atbash Cipher**. The repeating word patterns, single-letter word, and reverse alphabet substitution clearly support this conclusion. Although the Atbash Cipher is useful for learning classical cryptography, it provides very little security against modern cryptanalysis.

Result

Ciphertext → GSRH RH Z HVXIVG

↓

Apply Atbash Cipher (Reverse Alphabet)

↓

Recovered Plaintext → THIS IS A SECRET

Scenario 3: Secure Messaging System

Title

Keyword-Based Encryption Using the Vigenère Cipher

Introduction

To ensure secure communication within an organization, encryption techniques are used to protect confidential information from unauthorized access. One such classical encryption method is the **Vigenère Cipher**, which is a **keyword-based substitution cipher**. Unlike the Caesar Cipher, where the same shift value is applied to every letter, the Vigenère Cipher uses a **repeating keyword**, resulting in different shifts for different letters. This makes the encryption stronger and more secure than a fixed-shift cipher.

In this scenario, the employee encrypts the message "**HELLO**" using the keyword "**KEY**".

Given Data

Particular	Value
Plaintext	HELLO
Keyword	KEY
Encryption Technique	Vigenère Cipher
Cipher Type	Polyalphabetic Substitution Cipher

Theory

The **Vigenère Cipher** encrypts each letter of the plaintext by using the corresponding letter of a repeating keyword.

Encryption Process

Plaintext → Repeat the Keyword → Convert Letters to Numbers → Apply Encryption Formula → Ciphertext

Encryption Formula

Cipher Letter = (Plain Letter + Key Letter) mod 26

Where:

- A = 0
- B = 1
- C = 2
- ...
- Z = 25

(a) Step-by-Step Encryption Process

Step 1: Write the Plaintext

HELLO

Step 2: Repeat the Keyword

Since the keyword is shorter than the message, it is repeated until it matches the length of the plaintext.

Plaintext H E L L O

Keyword K E Y K E

Step 3: Convert Letters into Numerical Values

Plain Letter Value Keyword Letter Value

H 7 K 10

E 4 E 4

L 11 Y 24

Plain Letter Value Keyword Letter Value

L 11 K 10

O 14 E 4

\

Step 4: Perform Encryption

Using the formula:

Cipher = (Plain + Key) mod 26

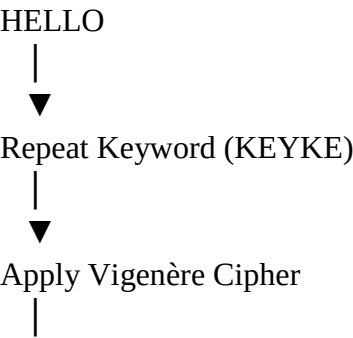
Plain Letter	Plain Value	Keyword Letter	Key Value	Calculation	Cipher Letter
H	7	K	10	$(7 + 10) \bmod 26 = 17$	R
E	4	E	4	$(4 + 4) \bmod 26 = 8$	I
L	11	Y	24	$(11 + 24) \bmod 26 = 9$	J
L	11	K	10	$(11 + 10) \bmod 26 = 21$	V
O	14	E	4	$(14 + 4) \bmod 26 = 18$	S

Encrypted Message

Plaintext Ciphertext

HELLO RIJVS

Encryption Flow



▼
RIJVS

(b) Influence of the Repeating Keyword

The keyword "**KEY**" is repeated as "**KEYKE**" to match the length of the plaintext.

Each letter of the keyword determines the shift applied to the corresponding plaintext letter.

Position	Plain Letter	Keyword Letter	Shift Applied	Cipher Letter
----------	--------------	----------------	---------------	---------------

1	H	K	+10	R
2	E	E	+4	I
3	L	Y	+24	J
4	L	K	+10	V
5	O	E	+4	S

Explanation

→ The keyword **KEY** is repeated as **KEYKE**.

↓

→ Each keyword letter provides a different shift value.

↓

→ The shift changes for every character instead of remaining constant.

↓

→ This produces a more secure ciphertext than the Caesar Cipher.

Because the shift value changes with every character, the ciphertext becomes more difficult to analyze using simple frequency analysis.

Advantages of the Vigenère Cipher

Advantage	Description
Stronger than Caesar Cipher	Uses different shifts for different letters instead of a single fixed shift.
Keyword-based encryption	A secret keyword increases security.
Reduces frequency analysis	Repeated letters in the plaintext may encrypt to different ciphertext letters.
Easy to implement	Suitable for learning classical cryptography concepts.

Conclusion

The message "**HELLO**" is successfully encrypted as "**RIJVS**" using the **Vigenère Cipher** with the keyword "**KEY**". The repeating keyword ensures that each character is encrypted using a different shift value, making the ciphertext more secure than a simple Caesar Cipher. This demonstrates how keyword-based encryption enhances confidentiality in secure communication.

Result

Plaintext → **HELLO**

↓

Keyword → **KEY** (Repeated as **KEYKE**)

↓

Apply Vigenère Cipher

↓

Ciphertext → **RIJVS**

Result: The plaintext "**HELLO**" is successfully encrypted as "**RIJVS**" using the **Vigenère Cipher**. The repeating keyword changes the shift for each character, providing stronger security than a fixed-shift encryption method.

4. Scenario 4: Suspicious Digital Asset Investigation

Title: Detection and Protection of Hidden Data Using Steganography

Introduction

During a digital forensic investigation, investigators may come across image files that appear completely normal but secretly contain hidden confidential information. This technique of concealing data inside digital media such as images, audio, or videos is known as **Steganography**. Unlike encryption, which makes data unreadable, steganography hides the very existence of the data. In this scenario, investigators suspect that a company logo contains hidden information and must determine whether any secret data has been embedded within the image.

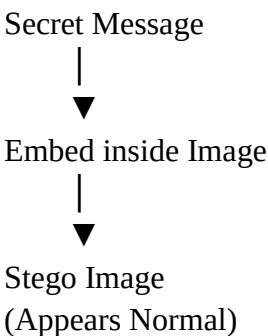
Given Scenario

Particular	Description
Digital Asset	Company Logo (Image File)
Suspicion	Hidden confidential information inside the image
Technique Involved	Image Steganography
Objective	Detect and secure hidden communication

Theory

Steganography is the process of hiding secret information inside another digital file without noticeably changing its appearance.

Working Principle



During forensic analysis, investigators examine the image using specialized tools to determine whether hidden information has been embedded.

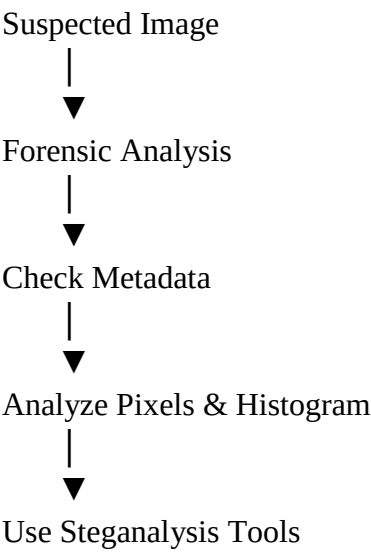
(a) Detecting Hidden Data in the Image

Investigators can use several forensic techniques to identify hidden information within an image.

Detection Techniques

Technique	Description
Steganalysis Tools	Software such as StegExpose or StegSecret can scan images for hidden data patterns.
Metadata Analysis	Examining the image metadata (file properties) may reveal suspicious modifications or hidden information.
Histogram Analysis	Comparing pixel color distributions can identify unusual changes caused by embedded data.
File Size Comparison	A larger-than-expected file size may indicate that extra hidden information has been added.
Pixel Analysis (LSB Analysis)	Examines the Least Significant Bits (LSB) of pixels, where hidden messages are commonly stored.

Detection Process





Explanation

Investigators begin by examining the image using forensic software. They compare the file size, inspect metadata, and analyze pixel values for irregularities. Many steganography techniques hide information in the **Least Significant Bit (LSB)** of image pixels, making the image appear visually unchanged while storing secret data. If unusual patterns are detected, investigators attempt to extract and analyze the hidden information.

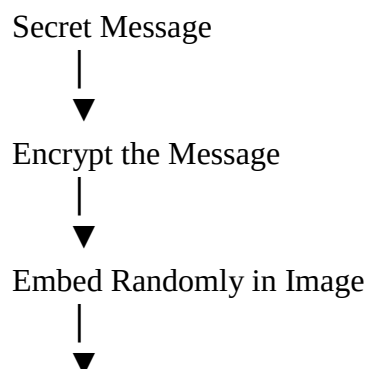
(b) Techniques to Improve Security of Hidden Communication

To prevent hidden data from being easily detected or extracted, stronger security measures should be used.

Recommended Security Techniques

Technique	Purpose
Encrypt the Secret Data Before Embedding	Even if the hidden data is extracted, it remains unreadable without the decryption key.
Randomized Pixel Embedding	Instead of hiding data sequentially, secret information is embedded randomly across different pixels, making detection much more difficult.

Security Enhancement Flow



Secure Stego Image

Advantages of Using These Techniques

Technique	Benefit
Encryption before embedding	Protects the message even if it is extracted.
Random pixel embedding	Reduces the chances of detection through statistical analysis.
Combined approach	Provides better confidentiality and stronger protection against forensic attacks.

Conclusion

Steganography allows confidential information to be hidden inside digital images without changing their visible appearance. During forensic investigations, techniques such as **metadata analysis, histogram analysis, steganalysis tools, and LSB analysis** help investigators detect hidden information. To improve the security of hidden communications, the secret message should first be encrypted and then embedded randomly within the image. This combination significantly reduces the risk of detection and unauthorized access.

Result

Image File



Forensic Analysis (Metadata + Histogram + LSB Analysis + Steganalysis Tools)



Detection of Hidden Data



Apply Security Enhancements

- Encrypt the secret message before embedding.
- Use randomized pixel embedding.

Result: Hidden information within the company logo can be successfully detected using forensic analysis techniques. Encrypting the secret message and embedding it randomly within the image provide stronger protection against detection and unauthorized extraction.

Scenario 5: Legacy Encryption in Corporate Communication

Title

Analysis of a Legacy Rail Fence Transposition Cipher

Introduction

Many older communication systems used **transposition ciphers** to protect confidential information. Unlike substitution ciphers, which replace letters with different letters, **transposition ciphers** rearrange the positions of the original letters without changing them. One of the most common transposition techniques is the **Rail Fence Cipher**.

In this scenario, the company uses a **3-Rail Rail Fence Cipher** to encrypt messages. The received ciphertext is "**WECRLTEERDSOEFEAOCAIVDEN**". The objective is to recover the original message, evaluate the weaknesses of this encryption method, and recommend a practical enhancement for modern secure communication.

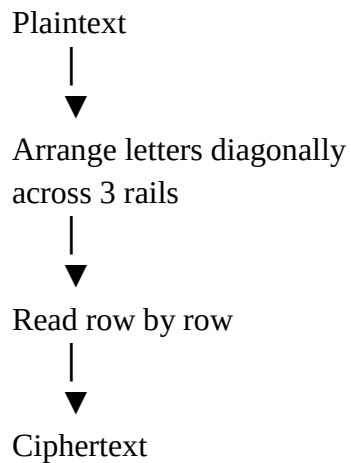
Given Data

Particular	Value
Ciphertext	WECRLTEERDSOEFEAOCAIVDEN
Encryption Technique	Rail Fence Cipher
Number of Rails	3
Cipher Type	Transposition Cipher

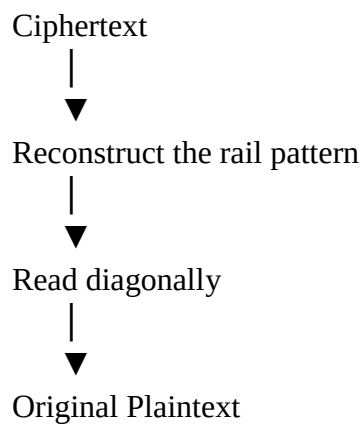
Theory

The **Rail Fence Cipher** encrypts a message by writing it diagonally across a fixed number of rails and then reading each rail row by row to produce the ciphertext.

Encryption Process



Decryption Process



(a) Determine the Original Message

Step 1: Given Ciphertext

WECRLTEERDSOEFEAOCAIVDEN

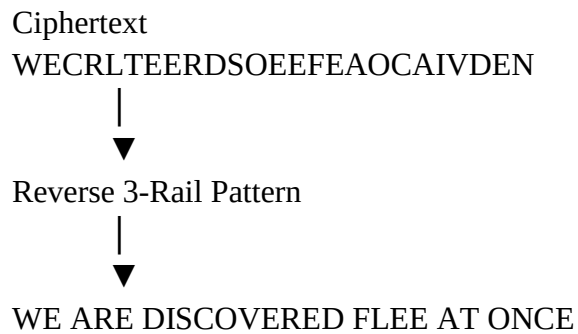
Step 2: Reverse the Rail Fence Arrangement

Using the **3-rail pattern**, the letters are placed back into their correct positions and then read diagonally.

Recovered Plaintext

Ciphertext	Recovered Plaintext
WECRLTEERDSOEFEAOCAIVDEN	WE ARE DISCOVERED FLEE AT ONCE

Decryption Flow



Explanation

The Rail Fence Cipher rearranges the positions of letters instead of replacing them. During decryption, the original zigzag (rail) pattern is reconstructed, and the letters are read diagonally to recover the original message.

The decrypted message is:

"WE ARE DISCOVERED FLEE AT ONCE"

which is a meaningful and grammatically correct sentence.

(b) Why is the Rail Fence Cipher Vulnerable?

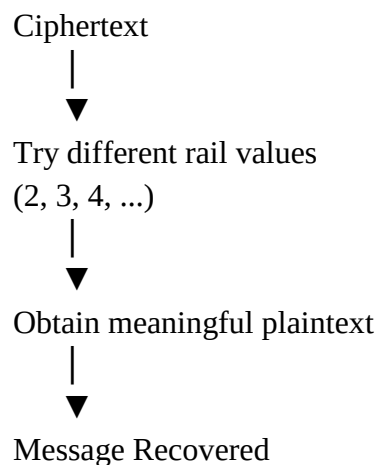
Although the Rail Fence Cipher provides basic message obfuscation, it is considered insecure for modern communication.

Limitations

Weakness	Explanation
Small key space	Only a few rail values are possible, making brute-force attacks easy.
No letter substitution	The original letters remain unchanged; only their positions are rearranged.
Vulnerable to pattern analysis	Attackers can reconstruct the zigzag pattern and recover the message.

Weakness	Explanation
No protection against modern attacks	It does not provide sufficient security for confidential corporate communication.

Why It Is Easy to Break



Because the number of possible rail arrangements is very small, an attacker can test each one until the correct plaintext appears.

(c) Practical Enhancement

Recommended Improvement

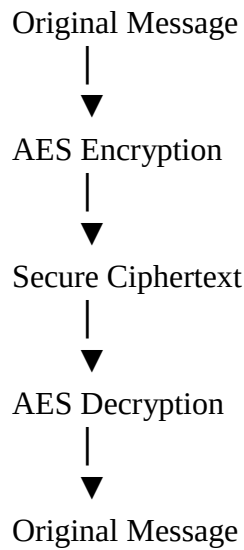
Use AES (Advanced Encryption Standard) instead of the Rail Fence Cipher.

AES is one of the most secure and widely used encryption algorithms in modern communication systems.

Benefits of AES

Feature	Benefit
Strong encryption	Protects data against brute-force and cryptanalysis attacks.
Secret key encryption	Only users with the correct key can decrypt the message.
High security	Used in banking, government, military, and cloud services.
Industry standard	Provides much stronger confidentiality than classical ciphers.

Improved Security Process



Conclusion

The ciphertext "**WECRLTEERDSOEFEAOCAIVDEN**" is successfully decrypted to "**WE ARE DISCOVERED FLEE AT ONCE**" by reversing the **3-Rail Rail Fence Cipher**. While this cipher demonstrates the basic concept of transposition-based encryption, it is highly vulnerable because attackers can easily reconstruct the rail pattern and recover the message. Replacing it with a modern encryption algorithm such as **AES** significantly improves the confidentiality and security of corporate communications.

Result

Ciphertext → **WECRLTEERDSOEFEAOCAIVDEN**

↓

Reverse 3-Rail Rail Fence Cipher

↓

Recovered Plaintext → **WE ARE DISCOVERED FLEE AT ONCE**

↓

Recommended Enhancement → Use AES Encryption for secure communication

Result: The original message was successfully recovered, the weaknesses of the Rail Fence Cipher were identified, and **AES encryption** was recommended as a practical solution for ensuring secure corporate communication.